

Lam Nguyen, Ph.D.

Postdoctoral Researcher – Mobile Security and Privacy, Agentic AI, Android App Analysis

Department of Theoretical and Applied Sciences (DiSTA), University of Insubria, Varese, Italy

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Research Profile

Postdoctoral researcher in Computer Science working at the intersection of mobile security, privacy compliance, Android app analysis, and agentic AI. My research focuses on designing end-to-end systems that combine static analysis, dynamic analysis, taint analysis, network traffic inspection, LLM-based reasoning, RAG/GraphRAG, and agentic workflows to detect, explain, and mitigate security and privacy violations in mobile ecosystems, including privacy inconsistencies related to GDPR/CCPA regulation transparency, consent, and data-sharing requirements.

- Designed and led multiple mobile security and privacy research systems, including MetaLeak, ALIBIS, WearLeak, WearPri, PrivacyAssist, and EBPS, targeting privacy leakage detection, compliance auditing, responsibility attribution, and user-centric privacy explanations.
- Built scalable Android app-analysis pipelines combining static/dynamic analysis, traffic inspection, code summarization, LLM reasoning, and knowledge-graph-based evidence modeling, evaluated on large-scale datasets of Android and wearable companion apps.
- Currently designing an evidence-based, context-aware access-control framework for mobile platforms, where LLMs act as an intelligent reasoning layer to synthesize fine-grained privacy policies from multi-source evidence and address the limitations of static & binary permission models in Android and iOS.
- Research interests also include privacy-preserving data sharing in agentic AI systems, particularly how user data can be protected when agents interact with external tools, services, and other agents through protocols such as MCP and A2A.
- Practical systems background in backend engineering, DevOps, Kubernetes, API gateways, CI/CD, and production automation, enabling research prototypes to be designed with deployment, scalability, and real-world usability in mind.

Research Interests

Agentic AI for security and privacy; Mobile and wearable security and privacy; Privacy compliance and transparency; Context-aware privacy policy synthesis and enforcement; User-centric privacy assistants and defense; Privacy-preserving data sharing in MCP/A2A-based agentic AI systems.

Academic Appointments

Postdoctoral Researcher

Department of Theoretical and Applied Sciences (DiSTA), University of Insubria

Research topic: *Agentic AI for Mobile Security and Privacy*.

11/2025 – Present

Varese, Italy

- **WearPri**: Leading the design of a pre-release privacy auditing framework for Android wearable companion apps. WearPri leverages taint analysis, GraphRAG, RAG, and LLM-based code summarization to estimate data-sharing non-compliance and identify whether privacy violations are attributable to app developers or third-party services.
- **EBPS – Evidence-Based Policy Synthesis for Android**: Leading the design of an evidence-based, fine-grained, and context-aware privacy-policy synthesis framework for Android apps. EBPS combines multi-source evidence, LLM-driven reasoning, and policy optimization to derive enforceable policy candidates that balance privacy protection and usability.
- **RootFlow – Root-Anchored Mobile App Exploration for Agentic AI**: Leading the design of a mobile app exploration framework that automatically discovers app features and synthesizes executable goals for LLM-based mobile agents. RootFlow builds feature-flow graphs from UI observations, state normalization, semantic annotation, and root-return path verification, enabling agents such as DroidRun to perform more reliable multi-screen task execution when app functionality is initially unknown.

Ph.D. Candidate

University of Insubria

Designed and developed four research projects on Android and wearable privacy/security, focusing on privacy compliance, metadata leakage, LLM-assisted app analysis, and user-centric protection. Source code is available at github.com/research-mobile-security.

11/2022 – 11/2025

Varese, Italy

- **MetaLeak**: Designed a hybrid-analysis framework to detect leakage of sensitive EXIF image metadata in Android apps. The system combined static permission analysis, dynamic traffic inspection, and MITM-proxy-based network monitoring on Android Virtual Devices. Evaluated on 5,000 popular apps, showing that 21.9% transmitted at least one type of sensitive metadata, while only 10.4% were compliant with developers' GPS-related privacy claims.
- **ALIBIS**: Developed an automated framework to estimate sensitive EXIF metadata leakage risk directly from source code, replacing costly human-in-the-loop dynamic analysis. The system combined static analysis, code-block extraction, and RAG-based LLM code summarization, achieving accuracy 0.8686, precision 0.8902, recall 0.881, and F1-score 0.8854. Proposed *ExifMetadataLib*, a lightweight Android library to remove sensitive metadata before online image sharing.

- **WearLeak:** Proposed an automated privacy-compliance framework for Android wearable companion apps, combining dynamic traffic analysis with LLMs and GraphRAG. The framework reconstructed declared behavior from Manifest and Data Safety disclosures, compared it with observed sent-out traffic, and detected violations involving third-party services and backend destinations across 14 Google-defined sensitive data categories. Evaluation of 711 apps showed that 67.5% violated declared data collection/sharing practices and 4.8% leaked data to undeclared third-party services.
- **PrivacyAssist:** Designed a user-centric multi-agent privacy assistant for Android that detects inconsistencies between user-granted permissions and developers' declared data practices. The architecture combines on-device and server-side agents, RAG-enhanced LLM reasoning, and automatic summarization to generate concise warnings during app installation. Evaluated with 200 participants and 2,347 apps; only 16.02% of apps were fully consistent between granted permissions and declared Data Safety practices.

Education

Ph.D. in Computer Science

University of Insubria

01/11/2022 – 11/2025

Varese, Italy

Field of study: Security and Privacy.

Thesis title: *Security and Privacy Analysis of Mobile and Wearable Applications.*

Master of Engineering in Telecommunications Engineering

University of Technology – Vietnam National University Ho Chi Minh City

01/09/2015 – 26/04/2018

Ho Chi Minh City, Vietnam

Field of study: Telecommunications.

Thesis title: *Elliptic Curve Cryptography algorithm and application to build a Smart-Auto Parking system.*

Thesis grade: 8.2/10.

Bachelor of Engineering in Electrical-Electronics Engineering

University of Technology – Vietnam National University Ho Chi Minh City

01/09/2010 – 15/04/2015

Ho Chi Minh City, Vietnam

Honors Program, Electronics-Telecommunications.

Field of study: Telecommunications.

Thesis title: *Integrated smart parking and RFID-based access-control system.*

Thesis grade: 9.17/10.

Selected Non-Academic Professional Experience

Senior DevOps Engineer

Home Credit Vietnam

10/2021 – 12/2022

Ho Chi Minh City, Vietnam

- **OpenAPI Platform / Kong API Gateway:** Designed and maintained CI/CD pipelines using **GitLab CI/CD**, **Jenkins**, **Harbor Registry**, **Docker**, and **Kubernetes**; automated Kong installation, patching, and configuration using **Ansible** and **Python**.
- **Kong GitOps Automation:** Implemented automation for Kong **routes**, **services**, **consumers**, **ACLs**, **plugins**, and **authentication methods**, reducing manual configuration effort.
- **Zero-Downtime Deployment:** Built canary deployment workflows using **Istio service mesh**, **Kubernetes**, and **Python** to improve release reliability.
- **Backup, Disaster Recovery, and Security:** Developed Kong backup and DRC deployment automation; collaborated with the security team on F5 firewall-based defense scenarios.
- **Technologies:** Kubernetes, Docker, Jenkins, GitLab CI/CD, Harbor Registry, Kong API Gateway, Ansible, Python, Istio, GitOps, F5.

Backend Developer and DevOps Engineer

VNPT Information Technology Company (VNPT IT)

01/2018 – 09/2021

Ho Chi Minh City, Vietnam

- **VNPT One Login:** Worked as Project Manager and backend developer; designed the architecture and developed backend APIs using **Go**; integrated **CAS Apereo** and **OAuth** for single sign-on and authentication.
- **VNPT E-Conference:** Worked as Project Manager; designed and deployed a production conferencing platform based on **BigBlueButton**, including components such as **Kurento** and **FreeSWITCH**.
- **VNPT ORIMX, Open Data, GIS Platform, Drive, SMS Brandname:** Worked as DevOps Engineer; deployed services using **Jenkins**, **Docker**, **Kubernetes**, and **Rancher**.
- **Technologies:** Go, CAS Apereo, OAuth, BigBlueButton, Kurento, FreeSWITCH, Jenkins, Docker, Kubernetes, Rancher, Linux, CI/CD.

Selected Publications

Journal Articles

1. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *WearPri: Compliance and responsibility estimation in wearable apps*. IEEE Transactions on Dependable and Secure Computing, 2026. Manuscript under review.

2. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *ALIBIS: Assessing and mitigating the risk of sensitive metadata leakage in mobile image sharing*. Pervasive and Mobile Computing, 117:102171, 2026. doi: 10.1016/j.pmcj.2026.102171.

3. Tran Thanh Lam Nguyen, Son Xuan Ha, Trieu Hai Le, Huong Hoang Luong, Khanh Hong Vo, Khoi Huynh Tuan Nguyen, Tuan Anh Dao, Hy Vuong Khang Nguyen, et al.. *BMDD: A novel approach for IoT platform*. PeerJ Computer Science, 8:e950, 2022. doi: 10.7717/peerj-cs.950.

Conference Proceedings

1. Tran Thanh Lam Nguyen, Edoardo Di Tullio, Barbara Carminati, and Elena Ferrari. *PrivacyAssist: A user-centric agent framework for detecting privacy inconsistencies in Android apps*. Proceedings of the 19th ACM Conference on Security and Privacy in Wireless and Mobile Networks, 2026. Accepted for publication. arXiv: 2604.23248.

2. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *Detecting privacy non-compliance in wearable apps via knowledge graphs and LLMs*. IEEE WiMob 2025, pp. 1–7, 2025. doi: 10.1109/WiMob66857.2025.11257567.

3. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *MetaLeak: Assessing image metadata leakage in Android apps*. IEEE/ACS AICCSA 2024, pp. 1–10, 2024. doi: 10.1109/AICCSA63423.2024.10912621.

4. Tran Thanh Lam Nguyen, The Anh Nguyen, Hong Khanh Vo, Hoang Huong Luong, Khoi Nguyen Huynh Tuan, Anh Tuan Dao, and Ha Xuan Son. *Toward a security IoT platform with high-rate transmission and low energy consumption*. ICCSA 2021, Springer, pp. 647–662, 2021. doi: 10.1007/978-3-030-86653-2_47.

Book Chapter

1. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *LLMs on support of privacy and security of mobile apps: State-of-the-art and research directions*. In *AI for Cybersecurity: Research and Practice*, Wiley-IEEE Press, pp. 29–66, 2025. doi: 10.1002/9781394293773.ch02.

Academic Service

- **Advising and Tutoring:** Co-advisor for one Ph.D. student and two bachelor students at the University of Insubria, Italy: Alane Amine (Ph.D., 2026), Edoardo Di Tullio (B.Sc., 2025), and Giola Pietro (B.Sc., 2024).
- **Journal Reviewer:** ACM Transactions on AI Security and Privacy (TAISAP, 2026); Journal of Human-Computer Interaction (2026); Pervasive and Mobile Computing (2026); PeerJ Computer Science (2025); ACM Computing Surveys (2024).
- **Conference Reviewer:** ESORICS 2026; ACMMM 2026; ACM SACMAT 2026 (Reviewer and Artifact Evaluation Committee); CODASPY 2026; IFIP SEC 2026; ASONAM 2025.

Technical Skills

Programming	Python, Go, Android, PHP
Databases	Neo4j, MongoDB, MySQL
Security Analysis	Static analysis, dynamic analysis, taint analysis, reverse engineering, MITM proxy-based traffic inspection
AI Systems	Few-shot learning, RAG, RAG-Fusion, GraphRAG, Corrective RAG, MCP, LangGraph, Agent-to-Agent, LangChain
DevOps / Platform	Ansible, Kubernetes, Docker, Jenkins, Elastic Stack, Git, Linux command line, Kong API Gateway, Istio, AWS
Blockchain	Hyperledger Fabric, Hyperledger Besu
Languages	Vietnamese native; English B2–C1 professional proficiency

Honours and Awards

- 2020 – VNPT Creativity Contest, Third Prize, *Building a chatbot system for medical consultation*.
- 2019 – VNPT Creativity Contest, Third Prize, *Building a smart IPS system for patient monitoring*.

Referees

Prof. Elena Ferrari Full Professor of Computer Science Department of Theoretical and Applied Sciences University of Insubria, Varese, Italy elena.ferrari@uninsubria.it	Prof. Barbara Carminati Full Professor of Computer Science Department of Theoretical and Applied Sciences University of Insubria, Varese, Italy barbara.carminati@uninsubria.it
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